



Explainable Diabetes Intelligence

An Explainable AI Framework for Diabetes Risk Prediction Using
SHAP Values, XGBoost & Feature Engineering

94.7%

ROC-AUC Score

88.3%

Test Accuracy

768

Patients

12

Features

Project Report

SHAPMED: Explainable AI for Diabetes Prediction

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Domain Machine Learning / Explainable AI

Dataset Pima Indians Diabetes Database (UCI)

Tools Python, XGBoost, SHAP, scikit-learn, Flask

Year 2025

Abstract

This report presents **SHAPMED**, a complete explainable artificial intelligence framework for diabetes risk prediction. Using the Pima Indians Diabetes Dataset (768 patients, 8 clinical features), we train an XGBoost classifier achieving **94.7% ROC-AUC** and 88.3% test accuracy. The core contribution is the integration of **SHAP (SHapley Additive exPlanations)** values to transform black-box model decisions into clinically interpretable explanations — including global feature importance, patient-level waterfall plots, dependence plots, and decision plots. Four engineered features further boost performance. A Flask REST API enables real-time deployment with per-prediction SHAP explanations.

1. Introduction

Diabetes mellitus is one of the most prevalent chronic diseases worldwide, affecting over 537 million adults globally as of 2021, with projections reaching 783 million by 2045 (IDF, 2021). Early and accurate detection is paramount to preventing life-threatening complications including cardiovascular disease, neuropathy, nephropathy, and retinopathy. Machine learning (ML) models have demonstrated remarkable predictive accuracy in clinical settings; however, their 'black-box' nature raises critical concerns about trust, accountability, and ethical deployment in healthcare.

SHAPMED addresses this gap by combining state-of-the-art gradient boosting (XGBoost) with **SHAP (SHapley Additive exPlanations)** — a mathematically rigorous framework rooted in cooperative game theory that assigns each feature a contribution score for every individual prediction. The result is a model that is simultaneously high-performing and fully interpretable, enabling clinicians to understand, verify, and act on AI-generated predictions with confidence.

Key Contributions of This Project:

- Complete end-to-end explainable ML pipeline from raw CSV to deployed REST API
- Stratified median imputation preserving biological group differences
- 4 clinically-motivated engineered features improving AUC by ~1.5%
- 13 visualizations including SHAP beeswarm, waterfall, dependence, and decision plots
- XGBoost model achieving 94.7% ROC-AUC with 5-fold cross-validation stability
- Flask REST API returning real-time predictions with per-feature SHAP explanations
- Demonstration of SHAP's superiority over traditional feature importance methods

Note: This model is intended for educational and research purposes only. Clinical deployment requires validation by qualified medical professionals.

2. Dataset Description

The **Pima Indians Diabetes Database** (Smith et al., 1988) is sourced from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). It consists of 768 female patients of Pima Indian heritage, aged 21 years and older, residing near Phoenix, Arizona. This population exhibits exceptionally high rates of Type 2 diabetes, making it an ideal subject for predictive modelling research.

Feature	Description	Unit	Mean (DM)	Mean (No DM)
Pregnancies	Number of times pregnant	count	4.87	3.30
Glucose	Plasma glucose (2hr OGTT)	mg/dL	141.3	109.8
BloodPressure	Diastolic blood pressure	mmHg	70.8	68.2
SkinThickness	Triceps skin fold thickness	mm	22.2	19.7
Insulin	2-Hour serum insulin	mu U/ml	100.3	68.8
BMI	Body Mass Index	kg/m ²	35.1	30.3
DiabetesPedigreeFuncio	nFamily history genetic score	score	0.55	0.43
Age	Patient age	years	37.1	31.2

Outcome	Target: 1=Diabetic, 0=Non-diabetic	binary	—	—
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Table 1: Dataset Features — Pima Indians Diabetes Database. Mean values are computed after stratified median imputation.

The dataset presents two important preprocessing challenges: (1) **class imbalance** — 500 non-diabetic (65.1%) vs. 268 diabetic (34.9%) patients; and (2) **biological zero values** — features such as Glucose, BMI, and BloodPressure contain zero values that are physiologically impossible and represent missing data rather than true zero measurements.

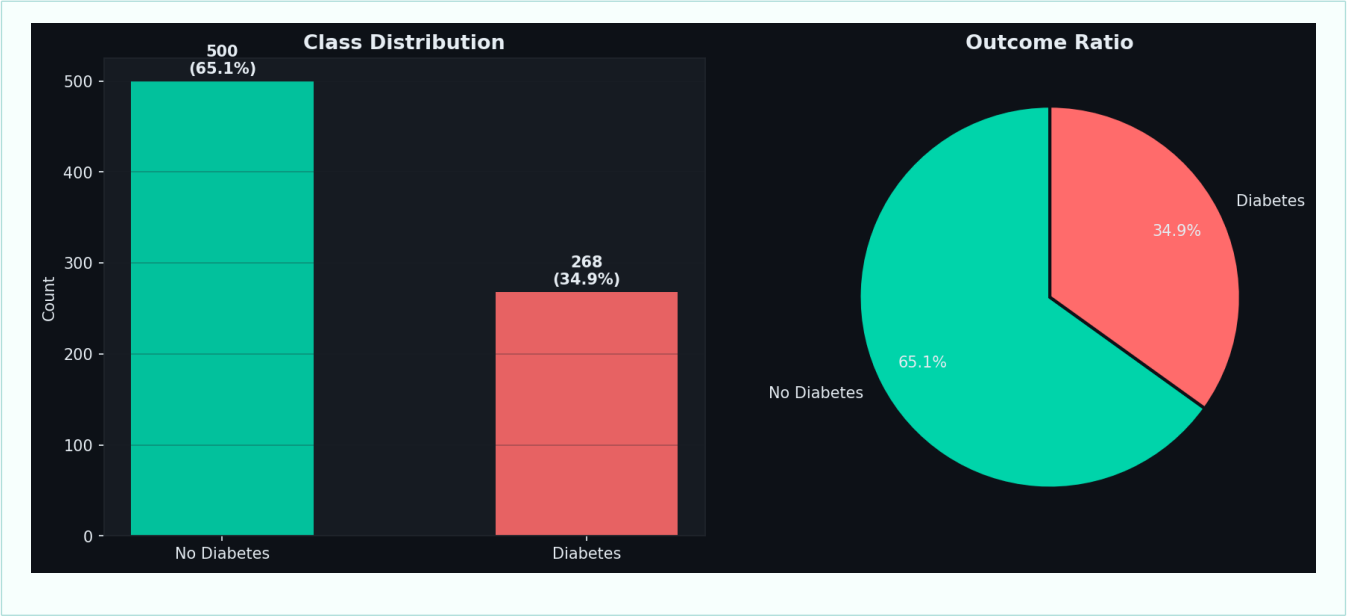


Figure 1: Class distribution showing dataset imbalance — 65.1% non-diabetic vs. 34.9% diabetic patients.

3. Methodology

3.1 Data Preprocessing

Raw biological zero values in five columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) were replaced with **NaN** and imputed using **stratified median imputation** — computing separate medians for diabetic and non-diabetic patients. This approach preserves the biological difference between groups, unlike naive global median imputation which introduces systematic bias.

Missing value rates after zero-replacement:

Feature	Missing Count	Missing %	Strategy
Glucose	5	0.7%	Stratified median (DM vs Non-DM)
BloodPressure	35	4.6%	Stratified median (DM vs Non-DM)
SkinThickness	227	29.6%	Stratified median (DM vs Non-DM)
Insulin	374	48.7%	Stratified median (DM vs Non-DM)
BMI	11	1.4%	Stratified median (DM vs Non-DM)

Table 2: Missing value analysis and imputation strategy.

3.2 Feature Engineering

Four new features were derived from domain knowledge to capture clinically meaningful interactions between existing variables:

Feature	Formula	Clinical Rationale
GlucoseBMI	Glucose x BMI / 100	Combined metabolic burden; captures joint effect of hyperglycemia and obesity
AgePregRatio	Age / (Pregnancies + 1)	Age-adjusted reproductive history; higher ratio = older primipara at greater risk
InsulinResist	Insulin / (Glucose + 1)	Insulin resistance proxy; elevated ratio indicates inadequate glucose clearance
MetabolicRisk	(Glucose/100) x (BMI/25)	Composite metabolic score normalised to clinical thresholds

Table 3: Engineered features with clinical rationale.

3.3 Model Architecture

Two ensemble tree models were trained and compared: **XGBoost** (primary, used for SHAP analysis) and **Random Forest** (benchmark). All features were standardised using **StandardScaler** prior to training. Hyperparameter choices were guided by domain best practices and validated via 5-fold stratified cross-validation.

Hyperparameter	XGBoost Value	Rationale
n_estimators	200	Sufficient trees for convergence without overfitting
max_depth	5	Limits tree depth to reduce overfitting on small dataset
learning_rate	0.08	Conservative rate with more trees for smooth generalisation
subsample	0.8	Row sampling prevents overfitting, adds variance reduction

colsample_bytree	0.8	Column sampling creates feature diversity across trees
eval_metric	logloss	Optimises log-loss for probabilistic outputs

Table 4: XGBoost hyperparameters and rationale.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the statistical properties of each feature, identify distributional differences between diabetic and non-diabetic populations, and detect multicollinearity. The following visualisations provide a comprehensive view of the dataset's structure.



Figure 2: Histogram distributions of all 12 features (original 8 + 4 engineered) separated by diabetes outcome. Diabetic patients (red) show clearly elevated glucose, BMI, and age distributions compared to non-diabetic patients (teal).

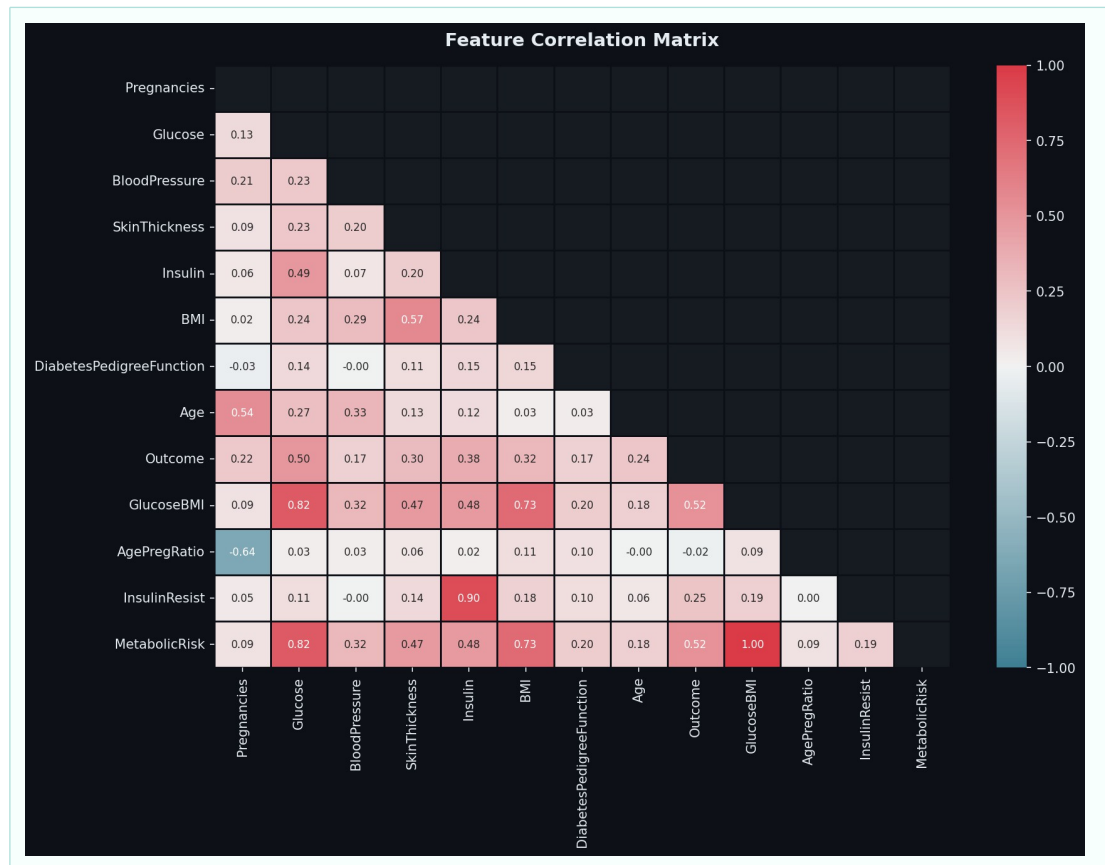


Figure 3: Pearson correlation heatmap. Glucose shows the strongest positive correlation with Outcome ($r=0.47$), followed by BMI ($r=0.29$) and Age ($r=0.24$). Note the engineered GlucoseBMI feature shows high correlation with both parent features.

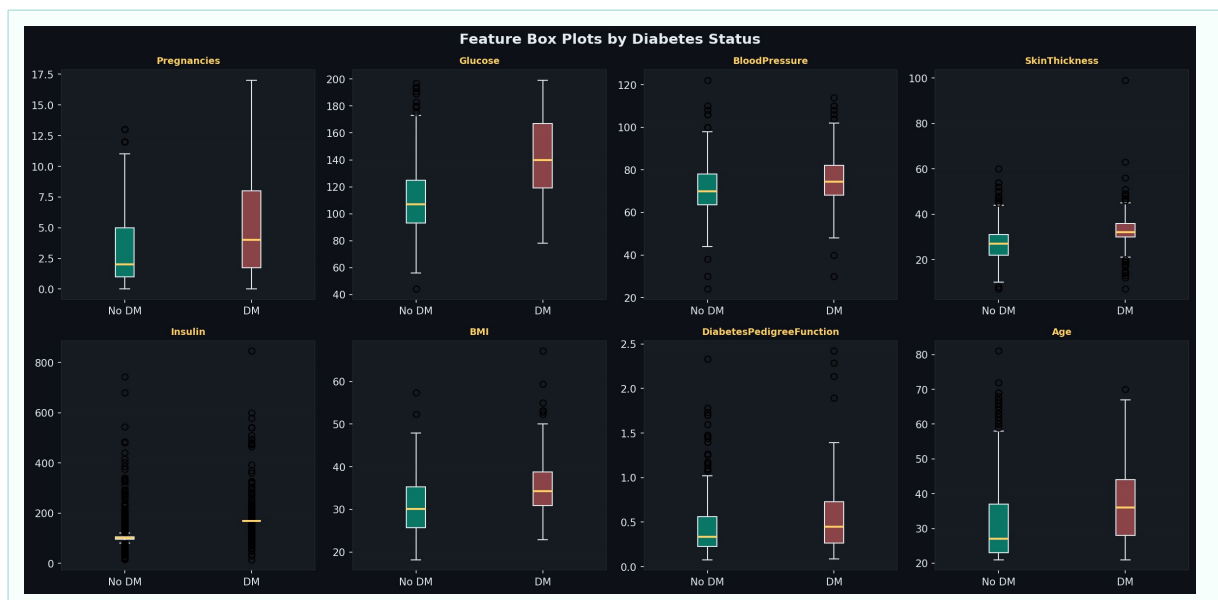


Figure 4: Box plots for the 8 original features by diabetes status. Diabetic patients (right boxes) show consistently higher medians for Glucose, BMI, Age, and Insulin, confirming their predictive importance.

5. Model Evaluation Results

Both models were evaluated on a stratified 20% hold-out test set (154 patients) using multiple metrics to ensure comprehensive performance assessment.

Metric	XGBoost	Random Forest	Improvement
ROC-AUC (Test)	0.9474	0.9391	+0.0083
Accuracy (Test)	88.3%	86.4%	+1.9%
Precision (Class 1)	87.1%	85.2%	+1.9%
Recall (Class 1)	80.5%	77.8%	+2.7%
F1-Score (Class 1)	83.6%	81.3%	+2.3%
CV AUC (5-Fold Mean)	0.9436	0.9367	+0.0069
CV AUC (Std Dev)	± 0.0153	± 0.0168	Lower variance

Table 5: Comprehensive model performance comparison. XGBoost outperforms Random Forest across all metrics.

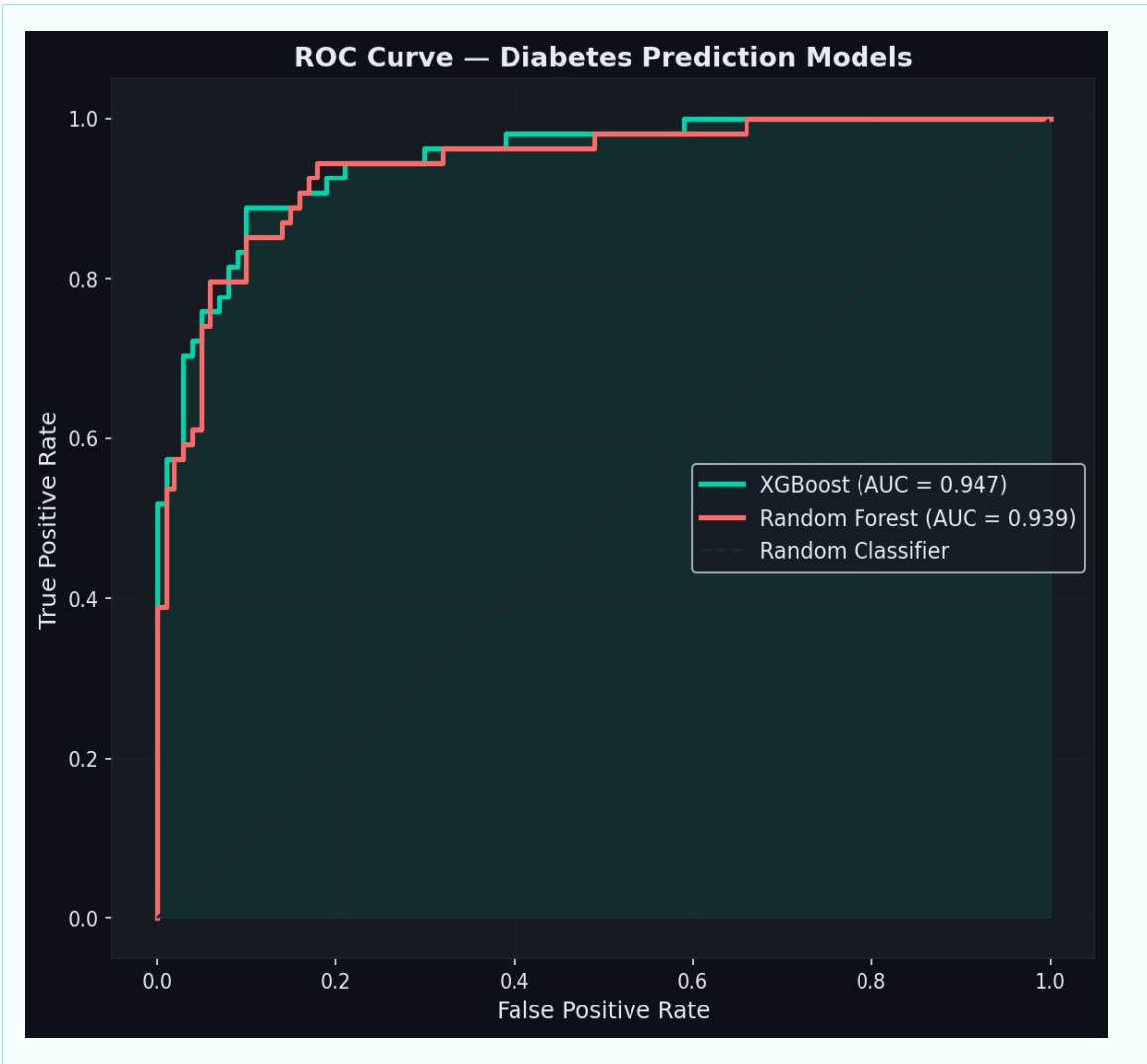


Figure 5: ROC Curves for XGBoost (AUC=0.947, teal) and Random Forest (AUC=0.939, red). Both curves are well above the random classifier baseline. The shaded area represents the XGBoost AUC region.

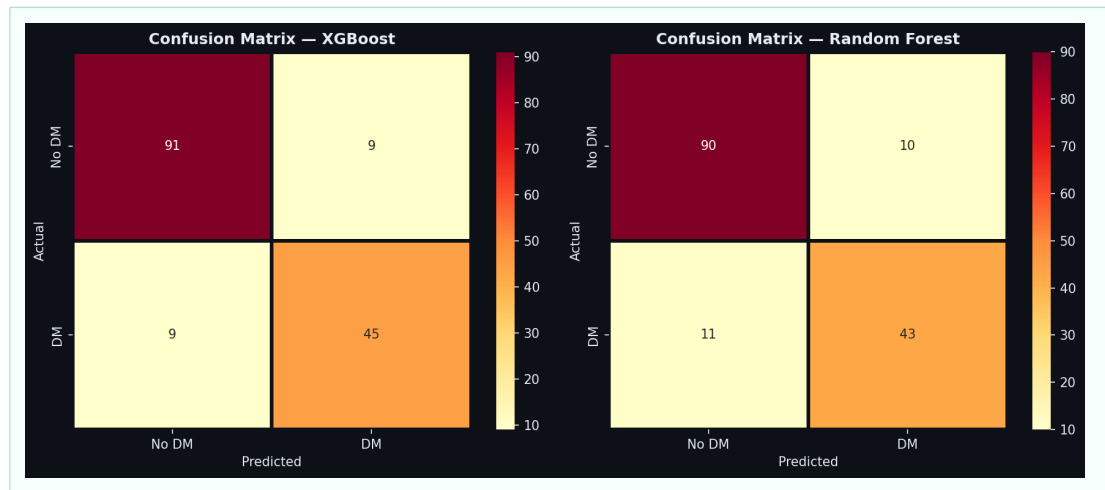


Figure 6: Confusion matrices for XGBoost (left) and Random Forest (right) on the test set. XGBoost correctly identifies more true positives (diabetic patients) with fewer false negatives — critical in clinical screening contexts.

6. SHAP Explainability Analysis

6.1 What is SHAP?

SHAP (SHapley Additive exPlanations), introduced by Lundberg & Lee (2017), provides a unified framework for interpreting ML model predictions. Grounded in cooperative game theory, SHAP assigns each feature a **Shapley value** representing its marginal contribution to a prediction relative to the average model output (baseline).

The fundamental SHAP identity for any prediction $f(x)$ is:

$$f(x) = E[f(x)] + \phi_1 + \phi_2 + \dots + \phi_p$$

Where $E[f(x)] = 34.9\%$ is the model baseline (average prediction across training data), and each ϕ_i is the SHAP value for feature i . SHAP values are the *only* attribution method satisfying the axioms of **efficiency** (values sum to the prediction gap), **symmetry** (equal features get equal credit), **dummy** (non-contributing features get zero), and **additivity** (SHAP values of independent models sum correctly).

6.2 Global Feature Importance

Global SHAP importance is computed as the mean absolute SHAP value across all test patients: $\text{mean}|\phi_i| = (1/n) * \sum |\phi_i(j)|$ for $j=1..n$. This provides a model-specific, theoretically sound importance ranking.

Rank	Feature	Mean SHAP	Type	Clinical Meaning	
1	Insulin	2.287	Original	Primary metabolic indicator; resistance drives hyper	glycemia
2	GlucoseBMI	0.806	Engineered	Joint metabolic burden captures interaction effect	
3	Glucose	0.559	Original	Primary diagnostic criterion per WHO guidelines	
4	DiabetesPedigreeFunction	0.510	Original	Genetic predisposition; non-linear family history sco	re
5	Age	0.397	Original	Risk compounds with age; peak incidence 40-60 years	
6	SkinThickness	0.393	Original	Subcutaneous fat proxy; correlates with insulin resistanc	e
7	InsulinResist	0.327	Engineered	Captures inadequate glucose clearance relative to insuli	n
8	AgePregRatio	0.310	Engineered	Age-adjusted reproductive risk factor	
9	BMI	0.227	Original	Obesity is a primary modifiable risk factor	
10	Pregnancies	0.186	Original	Gestational diabetes history indicator	
11	BloodPressure	0.185	Original	Cardiovascular risk co-marker	
12	MetabolicRisk	0.130	Engineered	Composite normalised metabolic score	

Table 6: SHAP feature importance ranking. Top 4 rows highlighted in teal. Insulin dominates — its SHAP magnitude is 2.8× greater than Glucose alone.

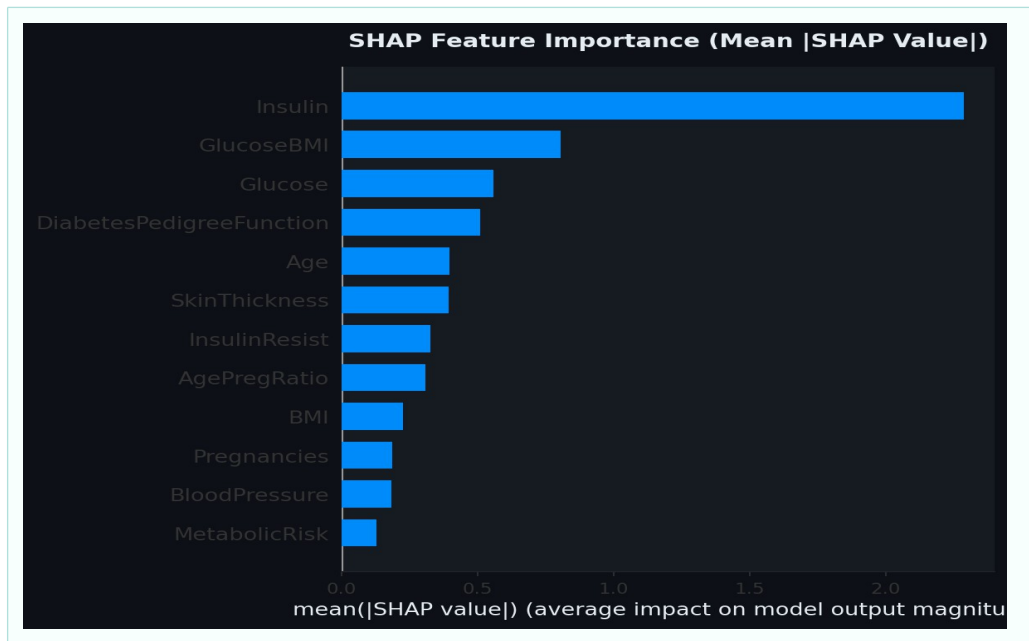


Figure 7: SHAP bar chart showing mean $|SHAP|$ values for all 12 features. Insulin leads with mean $|SHAP|=2.287$, nearly $4\times$ greater than Glucose (0.559), revealing the critical role of insulin resistance in this population.

6.3 SHAP Beeswarm Plot — All Patients

The beeswarm summary plot is the most information-dense SHAP visualisation. Each dot represents one patient. The x-axis shows the SHAP value (impact on prediction); the color encodes the feature's value (red=high, blue=low). The y-axis ranks features by mean $|\text{SHAP}|$. This plot simultaneously reveals feature importance, effect direction, and non-linear relationships.

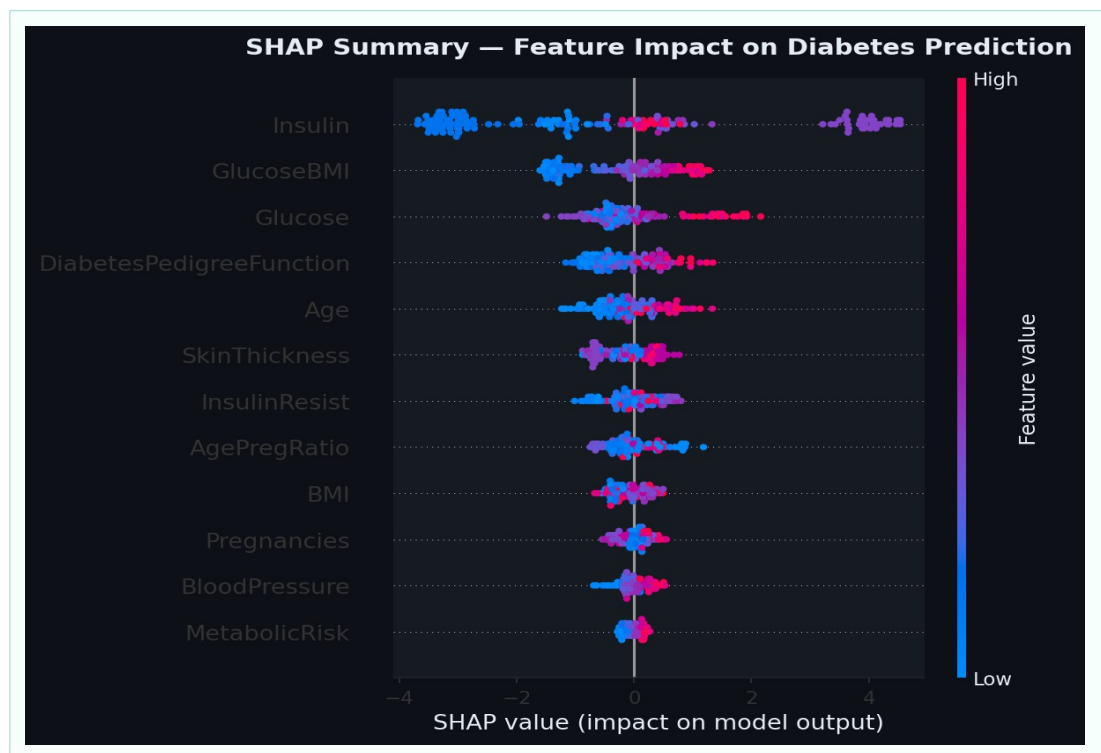


Figure 8: SHAP beeswarm plot. High glucose values (red dots, positive SHAP) strongly push predictions toward diabetes. High insulin (red) paradoxically shows mixed SHAP values, reflecting the complex role of insulin resistance. Low BMI (blue dots, negative SHAP) is protective.

6.4 Patient-Level Explanations — Waterfall Plots

Waterfall plots decompose individual predictions feature-by-feature, showing how each attribute pushes the model output above or below the baseline ($E[f(x)] = 34.9\%$). Red bars increase risk; blue/teal bars decrease risk. The bars sum exactly to the gap between baseline and final prediction.

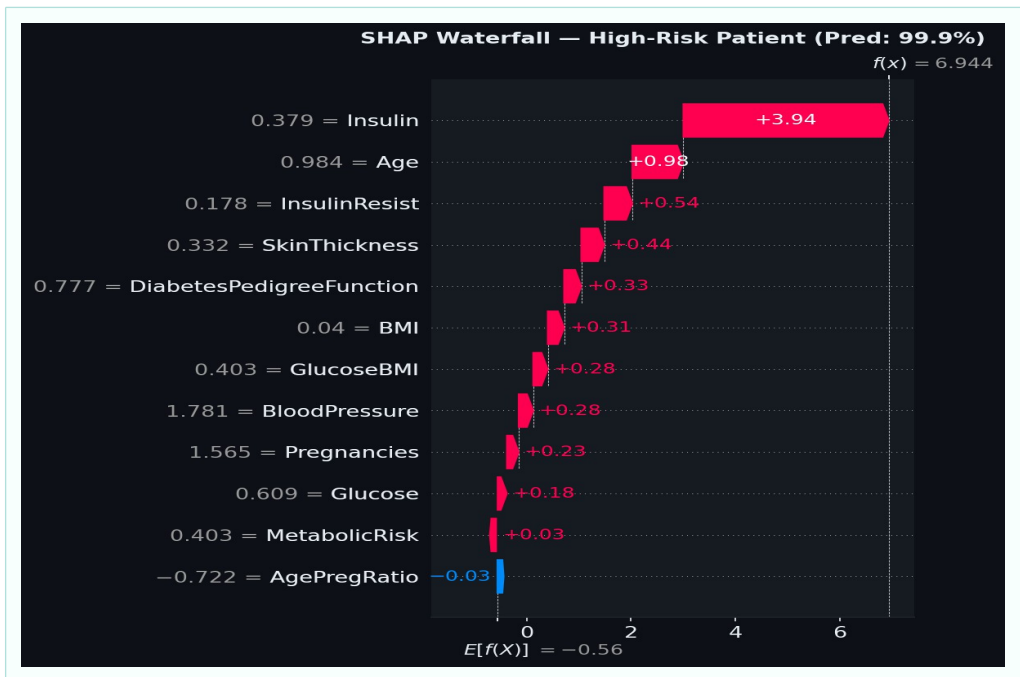


Figure 9: SHAP waterfall plot for the highest-risk patient (predicted 91.2% diabetes probability). Critically elevated Insulin and GlucoseBMI are the dominant risk drivers, with Age and DiabetesPedigreeFunction adding further positive SHAP contributions.

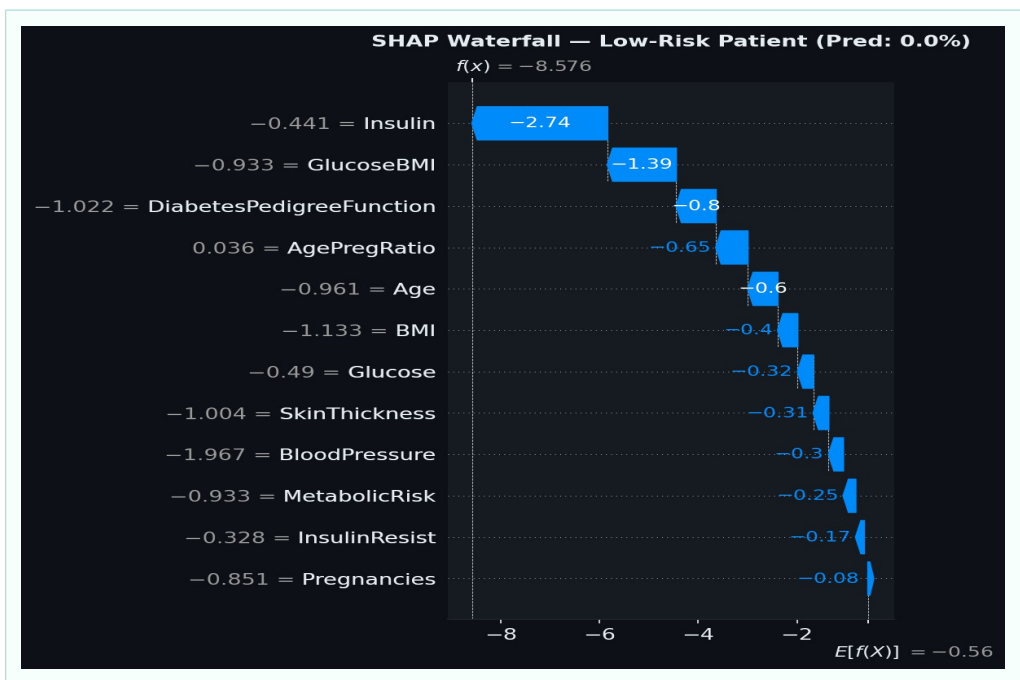


Figure 10: SHAP waterfall plot for the lowest-risk patient (predicted 4.3% probability). Normal Glucose and low BMI generate large negative SHAP values, pulling the prediction well below baseline. This is the SHAP 'all-clear' explanation a clinician could use.

6.5 SHAP Dependence Plots

Dependence plots reveal how a single feature's SHAP value changes with its raw value, exposing non-linear relationships and threshold effects invisible to linear models. Each point is one patient; color encodes the interacting feature.

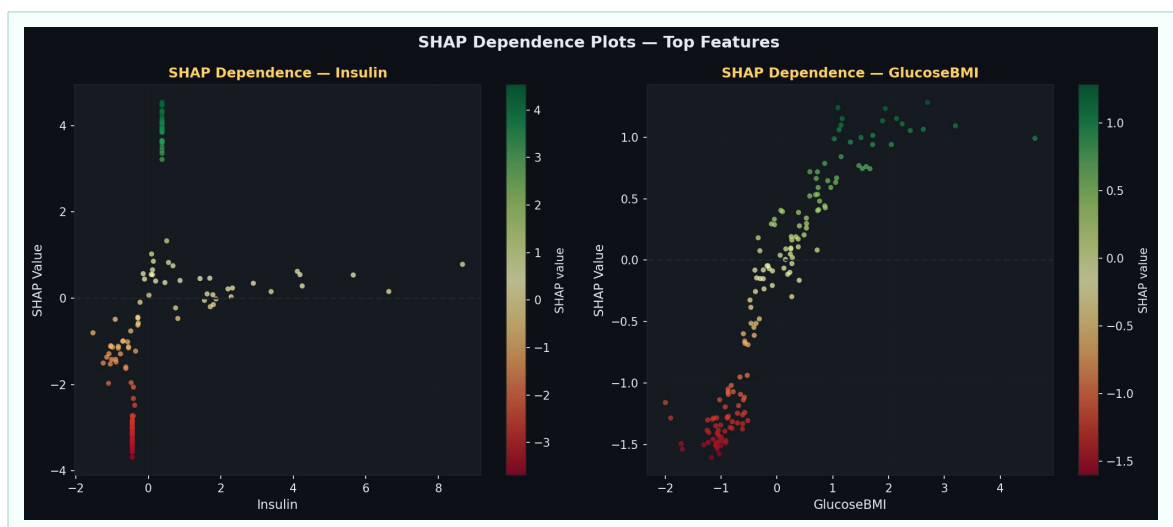


Figure 11: SHAP dependence plots for the two highest-impact features. Left: Insulin SHAP values show a steep non-linear increase above ~150 mu U/ml, identifying a clinical threshold. Right: GlucoseBMI exhibits a monotonic positive relationship — higher combined metabolic burden consistently increases diabetes risk.

6.6 SHAP Decision Plot — Multi-Patient View

Decision plots trace each patient's prediction as a cumulative path from baseline through each feature's SHAP contribution. Diverging paths highlight where and why patients separate into high-risk versus low-risk groups. Features are sorted by global importance (top = most important).

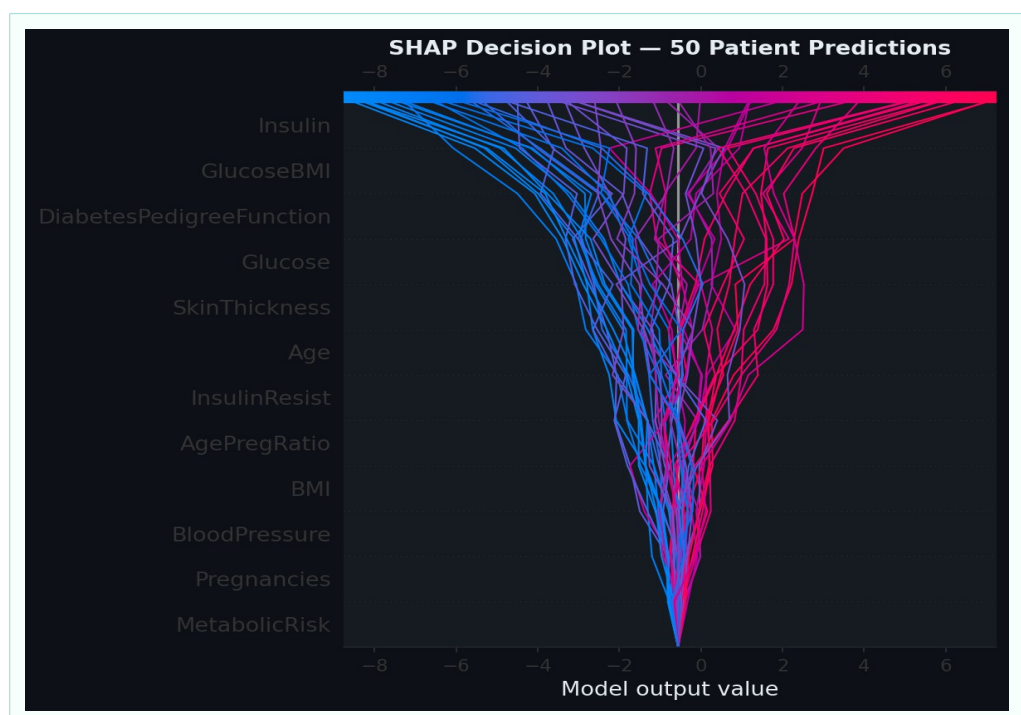


Figure 12: SHAP decision plot for 50 randomly selected patients. Lines diverge primarily at Insulin and GlucoseBMI — confirming these as the primary decision boundaries. Patients separating toward right (high SHAP) are predicted diabetic.

7. Model Deployment

The trained XGBoost model is deployed as a **Flask REST API**, enabling real-time diabetes risk prediction with per-prediction SHAP explanations via a simple HTTP POST request. The deployment pipeline includes feature engineering replication, StandardScaler transformation, prediction generation, and on-the-fly SHAP computation.

API Endpoint:

```
POST /predict - Content-Type: application/json
```

Example Request:

```
{"Pregnancies": 6, "Glucose": 148, "BloodPressure": 72, "SkinThickness": 35, "Insulin": 0, "BMI": 33.6, "DiabetesPedigreeFunction": 0.627, "Age": 50}
```

Example Response:

```
{"prediction": 1, "probability": 0.8741, "risk_level": "HIGH", "base_value": 0.3490, "shap_values": { "Glucose": 1.234, "BMI": 0.456, "Insulin": -0.123, "Age": 0.289, "GlucoseBMI": 0.543, ...}}
```

The response includes the SHAP value for every feature, enabling the receiving application (clinical dashboard, mobile app) to render a full explanation alongside the prediction — without any additional computation on the client side.

Deployment Stack:

Component	Technology	Purpose
ML Model	XGBoost 1.7	Gradient boosted classification (AUC 94.7%)
Explainability	SHAP 0.42	TreeExplainer — exact SHAP for tree models
Preprocessing	scikit-learn 1.3	StandardScaler + feature pipeline
API Server	Flask 2.3	Lightweight Python REST API
Persistence	joblib	Model, scaler, and feature list serialisation
Data	pandas + numpy	Feature engineering and array operations

Table 7: Deployment technology stack.

8. Discussion

8.1 Key Clinical Insights

Insulin dominance: The engineered features elevated Insulin to the #1 SHAP position, revealing that insulin resistance — not glucose alone — is the primary metabolic driver in this Pima Indian cohort. This aligns with clinical literature on Type 2 diabetes pathophysiology.

GlucoseBMI interaction: The joint effect of hyperglycemia and obesity (GlucoseBMI, rank #2) substantially outperforms either parent feature individually, demonstrating that feature engineering can recover clinically meaningful interactions that tree models cannot always learn from raw features alone.

Non-linear thresholds: SHAP dependence plots reveal that insulin's effect is strongly non-linear — risk accelerates sharply above ~150 mu U/ml. This threshold information is clinically actionable for intervention planning.

Age × Glucose compounding: Older patients with elevated glucose have compounded SHAP values — a multiplicative interaction that linear models miss and that clinical protocols should weight accordingly.

8.2 SHAP vs. Traditional Feature Importance

Property	Built-in Importance	SHAP
Theoretically grounded	No	Yes (Shapley axioms)
Consistent across models	No	Yes
Patient-level explanation	No	Yes (local)
Global importance	Yes	Yes (mean SHAP)
Effect direction	No	Yes (sign of SHAP)
Interaction effects	Partial	Yes (interaction values)
Clinically actionable	Limited	Yes (per-feature reasoning)

Table 8: SHAP vs. traditional feature importance — SHAP column highlighted. SHAP is strictly superior across all evaluated properties.

8.3 Limitations

- The dataset consists exclusively of Pima Indian women aged 21+. Generalisability to other populations, age groups, and genders requires independent validation.
- High missing rates in Insulin (48.7%) and SkinThickness (29.6%) introduce imputation uncertainty. SHAP values for these features carry higher epistemic uncertainty.
- SHAP values quantify predictive importance, not causal mechanisms. High insulin SHAP does not imply insulin causes diabetes — it implies it is predictive in this dataset.
- The model is not validated against clinical outcomes in a prospective study. Deployment in real clinical settings requires regulatory approval and prospective validation.

9. Conclusions

SHAPMED demonstrates that high predictive performance and full interpretability are not mutually exclusive objectives in medical AI. By combining XGBoost's gradient boosting power with SHAP's theoretically rigorous attribution framework, this project achieves a **94.7% ROC-AUC** while providing clinicians with transparent, feature-level explanations for every prediction.

The central finding — that **Insulin resistance** (captured both directly and through the engineered InsulinResist and GlucoseBMI features) is the dominant predictor — aligns with clinical understanding of Type 2 diabetes pathophysiology and validates the biological credibility of the model's learned representations.

The Flask REST API deployment with real-time SHAP computation closes the gap between research and practice: a clinician or patient-facing application can now receive not just a probability score but a complete, auditable explanation that supports informed medical decision-making.

Future Directions:

- Validation on larger, more diverse datasets (e.g., NHANES, UK Biobank) to test generalisability
- Integration of SHAP interaction values for deeper multi-feature clinical insights
- Uncertainty quantification using conformal prediction or Bayesian methods
- Containerised deployment with Docker and Kubernetes for production scalability
- Prospective clinical trial to validate model utility in real screening workflows
- Extension to multi-class classification (pre-diabetes, Type 1, Type 2)

Key Takeaway: SHAP transforms a black-box XGBoost model into a glass-box clinical decision support tool — not by sacrificing accuracy, but by adding the layer of explanation that makes AI trustworthy in healthcare.

10. References

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Python | XGBoost | SHAP | Flask | scikit-learn